

Exercise 09 for MA-INF 2201 Computer Vision WS25/26
25.01.2026
Submission deadline on 01.02.2026

1. Extract the intrinsic camera parameters along with the distortion parameters using a checkerboard-based camera calibration. For this task you have to calibrate your own camera using your own photos of a checkerboard (save the images in a folder camera calibration). Print the results of the calibration in a readable way and compare i.e. the focal length with the given camera specifications. To check the calibration compute the mean re-projection error considering the real scale of objects. You can use the predefined OpenCV functions for the task. (4 points)

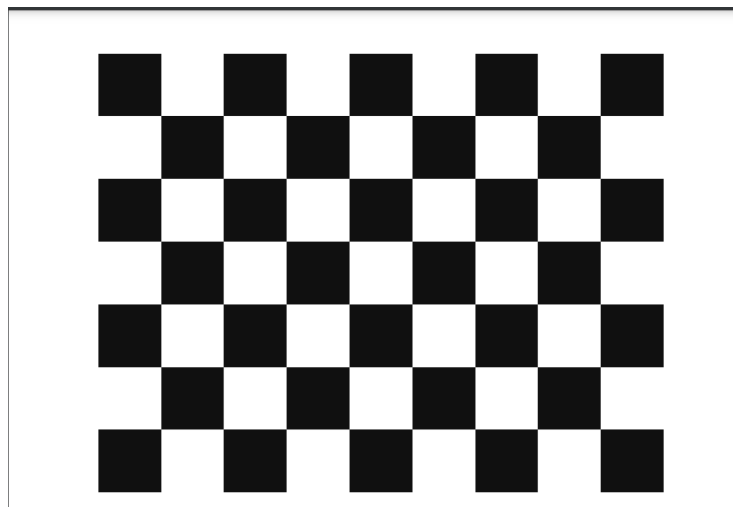
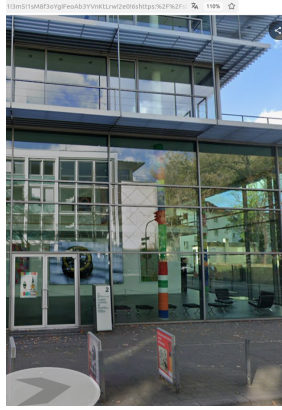


Figure 1: Example of checkerboard for camera calibration.

2. The folder data contains three images (A.png, B.png, C.png):
 - (a) Considering the image pair (A, B) pick at least 6 identical points on the building facade (Figure 3 highlighted in red) in both images. Implement an interactive selector to pick these points. (1 point)
***Important:** To allow easier grading **comment out** the interactive selection and directly provide the selected points coordinates.*
 - (b) Compute the homography transformation $\mathbf{H}_{A \rightarrow B}$ using homogeneous coordinates. Compute the errors between the transferred and measured points and print the alignment errors. Use your **own implementation** following the lecture notes to compute the $\mathbf{H}$ transformation. (2 points)
 - (c) Use the transformation $\mathbf{H}_{A \rightarrow B}$ to map the source image into the geometry of the target image, thus visualize and store the result as in Figure 3a. For this task you can use the function `cv2.warpPerspective`. (1 point)
 - (d) Repeat steps (a) and (b) for the image pair (B, C) to compute $\mathbf{H}_{B \rightarrow C}$. (1 point)
 - (e) Derive the $\mathbf{H}_{A \rightarrow C}$ transformation and, together with the previously computed homographies, warp A B and C to get a result like in Figure 3b. (3 points)



(a) A.png



(b) B.png

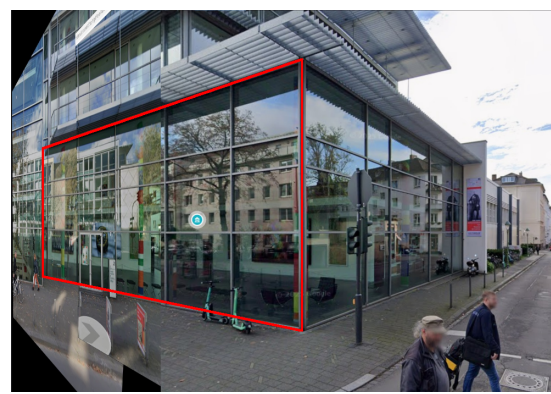


(c) C.png

Figure 2: Task 2 input images in data folder.



(a) Warping of A and B.



(b) Warping of A B and C together.

Figure 3: Expected results for task 2.

3. The folder data contains three images (Fuji_1.png, Fuji_2.png, Fuji_3.png):

- (a) Use the SIFT algorithm to extract key-points and their corresponding descriptors for the images. You can use the class SIFT from OpenCV. *(0.5 points)*
- (b) Implement the best match ratio test with the threshold 0.3 as follows: Compute the square of the Euclidean distance between all pairs of key-points across the two images. For each key-point, find the most similar and the second most similar key-point from the other image. Only keep key-points which are consistent in both ways and pass the ratio test. Display your matching key-points using the function `cv2.drawMatchesKnn`. *(1.5 points)*
- (c) Assume you don't have any prior knowledge about the images, this means that you can't decide beforehand which of the three to consider as reference image. Compute matches between all image pairs and base the reference image choice on this extracted information. *(2 points)*
- (d) Implement RANSAC to compute the homography transformations $\mathbf{H}$ between the detected reference image and the other two. You can use the function `cv2.PerspectiveTransform` to calculate the projective transformation and `cv2.warpPerspective` to warp the image. *(3 points)*

- (e) Stitch the three images together to obtain an image mosaic similar to the one in Figure 5. (1 point)

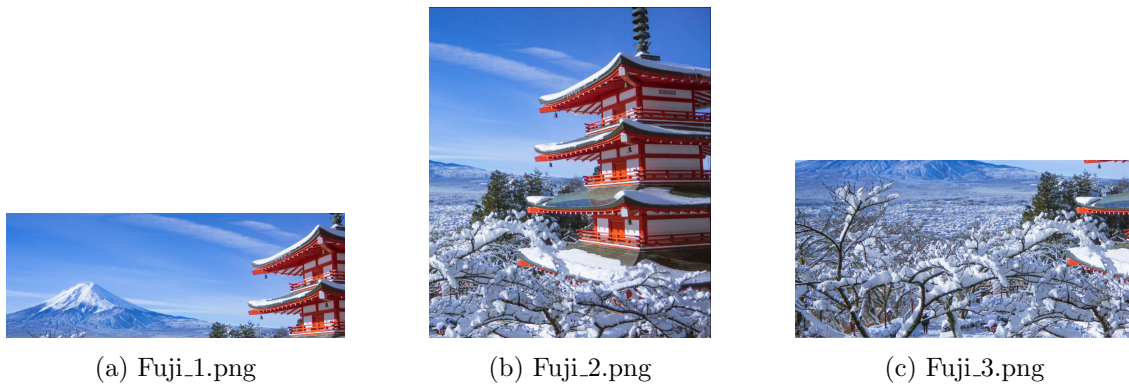


Figure 4: Task 3 input images in data folder.



Figure 5: Expected result for task 3.

How to report on this assignment. Please exclude the data folder from your submission. Use the already provided template `.py` files, but feel free to change the code structure as you want.

Questions If you have questions, find any ambiguities/mistakes, please contact: Enrico Pallotta (pallotta@iai.uni-bonn.de).

Important: You are **not allowed** to use any additional python modules beyond the ones imported in the template. Otherwise, you will not get any points.

Grading: Submissions that generate runtime errors or produce obviously useless results (i.e., nans, inf or meaningless visual outputs) will receive at most 50% of the points.

Plagiarism: Plagiarism in any form is prohibited. If your solution contains code directly copied from any source (i.e., other students), you will receive **0 points** for the entire exercise sheet.

Submission: You can complete the exercise in a group of two, but only one submission per group is allowed. Include a `README.txt` file with your group members in each solution. Points for solutions without a README file will only be given to the uploader.